

Chapter 9

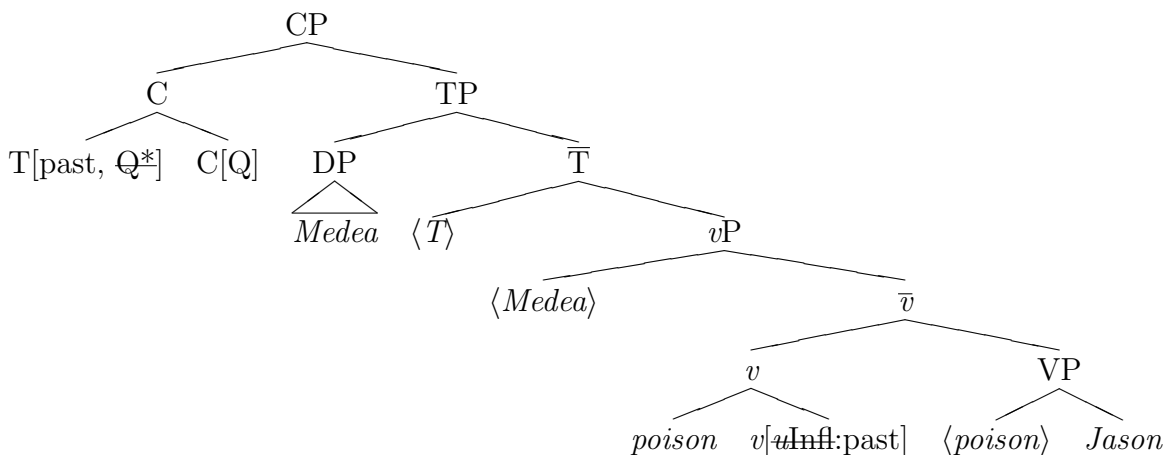
Wh Movement

9.1 Introduction

In the previous chapter we saw that clauses were headed by a functional projection which we labelled C. We saw evidence for C in matrix questions, as well as in embedded clauses. This evidence came from the phenomenon of Subject Auxiliary Inversion, which we analysed as movement of C to T driven by a strong $[Q^*]$ feature which is the value given to an uninterpretable $[u\text{clause-type}]$ feature on T. If T bears $[Q^*]$, then it moves into a local configuration with C:

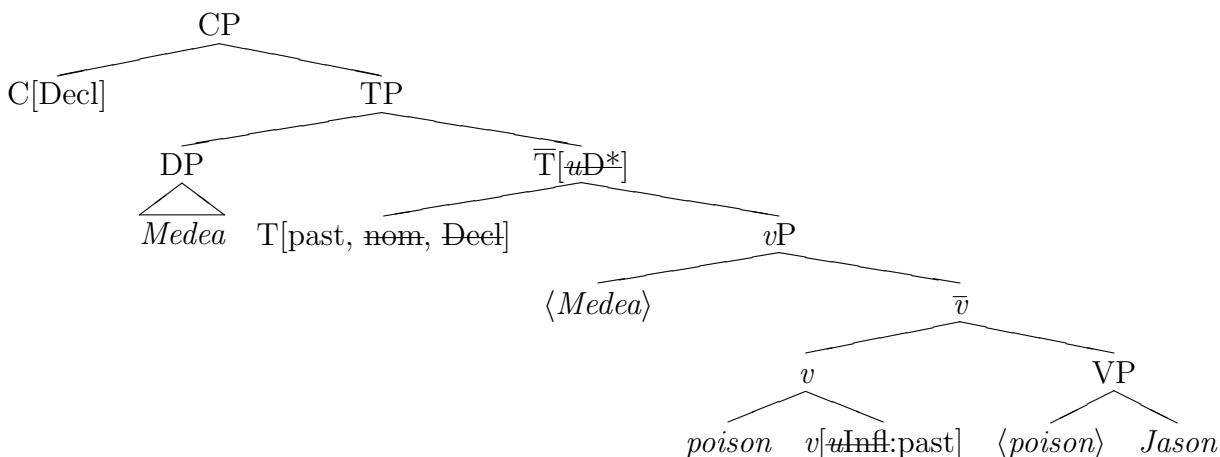
(1) Did Medea poison Jason?

(2)



If T's $[u\text{clause-type}]$ feature is valued as [Decl], then it is weak, and so T does not move to C. We then just get a simple declarative clause, with the following structure:

(3)



Recall that, by the PTR, v bearing checked tense features is not pronounced when the chain between T and v is broken, hence we have *poison* rather than *poisoned*.

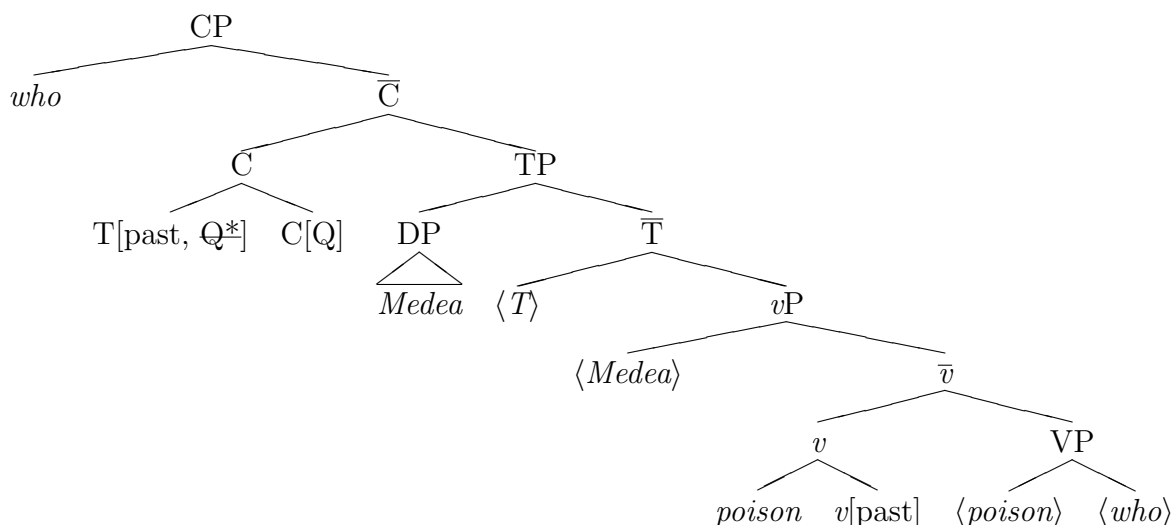
This gives us evidence for C as a head position, but we might also like to see evidence that C has a specifier. We saw some evidence from the V2 phenomenon found in Germanic that C may project a specifier, but evidence is also closer at hand in English constructions like the following sentence:

- (4) Who did Medea poison?

This kind of sentence is known as a **wh-question**, and has some interesting properties which we will investigate in this chapter and the next.

On the assumption that the tree in (2) provides the right analysis for cases where the auxiliary precedes the subject, we then ask the question what the structural position of *who* in (4) is. On analogy with the structure of TP, vP and DP, we hypothesize that *who* is in the specifier of CP.

- (5)



In this tree, we have moved *who* from the complement of V position to the specifier of TP. We will look at the way that this movement operation works in more detail in the

remainder of this chapter. At a first approximation, we notice that *who* in (4) is assigned the Theme θ -role by *poison*; the question in (4) is a question about the person who is poisoned. Given the UTAH, this suggests that *who* is initially Merged with *poison* in the usual Theme θ -position and that it subsequently moves to its surface position, as diagrammed in (5).

This analysis predicts that if we take the tree in (5) and embed it under a verb which selects a question (giving an embedded question), then *who* will appear immediately between the selecting verb and the complementizer of the embedded clause. This prediction is almost correct, but not quite, as we can see from (6):

- (6) I asked who Medea poisoned.

The sentence in (6) has *no* embedded complementizer. Contrast this with (7):

- (7) I asked if Medea poisoned Jason.

It appears that the embedding complementizer which occurs with wh-movement in standard English is null, unlike the embedding complementizer that appears with simple embedded yes/no questions. That this is the right approach is backed up by dialects like those spoken in Northern Ireland, which allow embedded inversion for yes/no questions:

- (8) I asked did Medea poison Jason.

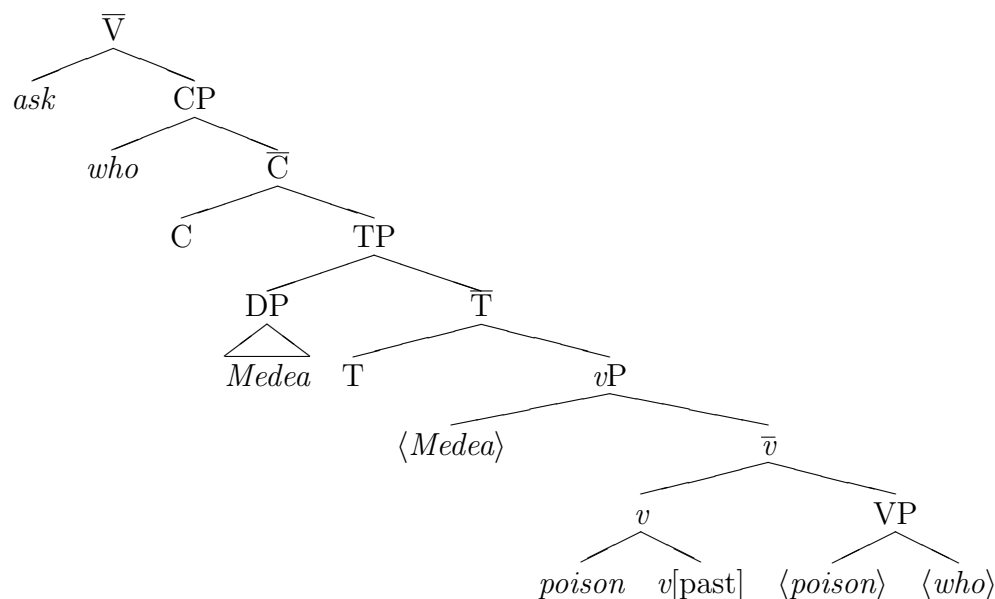
These dialects also have inversion in embedded wh-questions:

- (9) I asked who did Medea poison.

Since, by hypothesis, the auxiliary is in the C position, the word *who* is sandwiched between the selecting verb and the auxiliary, as expected.

Structurally, then, we assume that an embedded wh-question looks as follows:

- (10)



In the remainder of this chapter we will explore the wh-construction further, determining what the properties of the items that undergo this movement are, what motivates this approach, and how we capture that particularities of the construction in our system.

9.2 Elements that undergo wh-movement

In our example of a wh-question, we suggested that a movement operation applies to the word *who*, displacing it from its θ -position so that it ends up in the specifier of CP. This operation does not apply to just any word or phrase in English, but only to a specific subset of words and phrases. Compare, for example, (11) and (12):

- (11) Who did Medea poison?
- (12) *Someone did Medea poison.

The expressions to which this movement process applies are called **wh-expressions**, because most of them, in English, begin with the letters *wh*.

- (13) What have you eaten?
- (14) When did you arrive?
- (15) Where are you living?
- (16) Which book are you reading?
- (17) Why are you leaving?
- (18) How are you feeling?

Only the word *how* does not begin with *wh*. Of course, wh-movement does not apply to just any word that begins with *wh*; this movement operation isn't triggered by phonological features, but, just like all of our other cases of Movement, it's triggered by morphosyntactic features:

- (19) *Whales have I seen.
- (20) *Whiskey do I drink.

We therefore set up a morphosyntactic feature [wh] which picks out this class of elements. We will assume that [wh] is simply a privative (that is atomic) feature, which does not participate in feature valuing under Agree.

9.2.1 Wh-expressions

Most of our wh-expressions are simple words: *when*, *where*, *who*, *what* and so on. These words are just like [wh] versions of the pro-forms: *then*, *there* and the third person pronouns. We treat them, then, as heads bearing the appropriate category feature together with [wh]:

- (21) where [wh, P, locational]
- (22) when [wh, P, temporal]

(23) who [wh, D, human]

(24) what [wh, D, non-human]

I have used various interpretable semantic features to distinguish these wh-forms yet further. Syntactically, *where* and *when* consist of the same features. Note that the first three of these expressions are pronominal in nature. They cannot combine with another syntactic object and project a larger wh-phrase:

(25) *[where place] are you living.

(26) *[when time] will you be there.

(27) *[who guy] did you see.

Under the theory we developed in chapter 4, this means that they are both lexical items, and, simultaneously, phrasal items. They are initially Merged as arguments or adjuncts, and as such, they project no further. They are, at this point of the derivation, phrasal. They then undergo movement as phrases, and so target the specifier rather than the head of CP.

The wh-forms *what* and *which* do allow a complement though. Both can combine with nominal phrases to form a larger wh-DP:

(28) [Which poem about Achilles] did Homer recite?

(29) [What kind of actor] is he?

These words then seem to be the [wh] counterparts of determiners, or demonstratives:

(30) Homer recited [the poem about Achilles]?

(31) He is [that kind of actor]

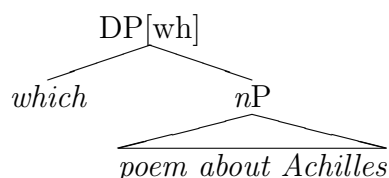
This idea is further backed up by the fact that, in English, these wh-elements are in complementary distribution with Ds:

(32) *Which the poem did Homer recite?

(33) *What a kind of actor is he?

We will therefore treat them as *wh-determiners* which combine with *nP* complements:

(34)



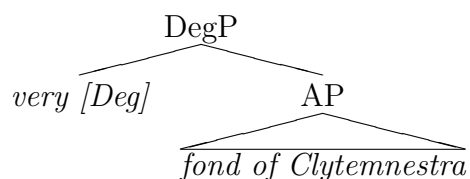
The wh-forms *how* and *why* don't have such clear non-wh pro-form counterparts (although the somewhat archaic *thus* provides a partial parallel), but syntactically they seem to behave in a similar way to *where* and *when*. *How* is also capable of combining with adjectives and adverbs to create larger structures:

(35) [How fond of Clytemnestra] is Agamemnon?

(36) [How quickly] did the Greeks take Troy?

We will treat these wh-words as being wh-versions of degree expressions like *very*, and for concreteness, we will assume that these expressions are of a category Deg(ree):

(37)



Deg is a functional head which appears above APs and AdvPs, in much the same way that D projects above *n*Ps and T above *v*Ps. This means that we need to state a new Hierarchy of Projections:

(38) **Hierarchy of Projections:**

Clausal: C › T › (Neg) › (Perf) › (Prog) › (Pass) › *v* › V

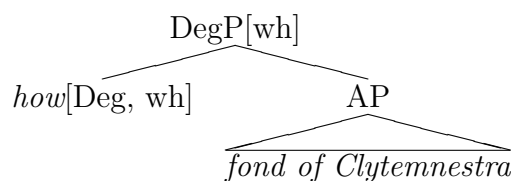
Nominal: D › (Poss) › *n* › N

Adjectival (Deg) › A

We will not pursue the functional structure dominating adjectival elements any further here. See the reference section for further reading.

With this in place, we can now treat *how* as the wh-version of this Deg head:

(39)



Other languages also display morphological regularities in their wh-words. I have given below the wh-words of Zuni, a Native American language of New Mexico, of Kiowa, an unrelated Native American language spoken in Oklahoma, and Classical Hebrew, a Semitic language. None of these languages is related to the Indo-European group to which English belongs, but all display regularities in the construction of their wh-expressions.

(40)

English:	who	what	where	when
Zuni:	cop	kwap	hop	kyaayip
Kiowa:	hà:tèl	hò:ndé	hà:yá	há:oy
Hebrew:	mi	mah	‘ayin	matay

As you can see, all of the Zuni *wh*-expressions end in *-p*; all of the Kiowa expressions begin with an *h* and three of the four Hebrew expressions begin with an *m*. Of course, not all languages display morphological regularities in their *wh*-expressions (Turkish, for example, does not), however, enough unrelated languages do to suggest that the feature [wh] is morphologically motivated as one of the features that UG makes universally available.

9.2.2 Interpretation of *wh*-questions

When we turn to the semantics of these expressions, we find that they have a fairly uniform interpretation. It is usually possible to paraphrase *wh*-questions as though they contain an **interrogative quantifier**. By interrogative quantifier, I mean that the question can usually be paraphrased as ‘For which *x* is it the case that...’. Here are some examples:

(41) **Who** ate the cake?

For which *x*, *x* a human, is it the case that *x* did that?

(42) **When** did you eat the cake?

For which *x*, *x* a time, is it the case that you ate the cake at time *x*.

(43) **Why** did you eat the cake?

For which *x*, *x* a reason, is it the case that you ate the cake for the reason *x*.

(44) [**Which** girl] ate the cake?

For which *x*, *x* a girl, is it the case that *x* ate the cake.

(45) **What** did she eat?

For which *x*, *x* a thing, is it the case that she ate *x*.

(46) **Where** has he put the cake?

For which *x*, *x* a place, is it the case that he put the cake in place *x*.

(47) **How** did you eat the cake?

For which *x*, *x* a manner of doing something, is it the case that you ate the cake in manner *x*.

(48) [**How** quickly] did you eat the cake?

For which *x*, *x* a measure of quickness, is it the case that you ate the cake at that measure of quickness.

Recall that in our discussion of the motivations behind positing features we said that the strongest motivation for a feature was when there was some morphological and semantic expression of that feature. This is exactly the situation here: we have a limited class of words, most of which share a morphological peculiarity, and all of which share a semantic

commonality. Moreover, this class of words displays the same syntactic behaviour. They all may move to the specifier of CP.

We are justified then in setting up the [wh] feature which picks out this class of elements.

9.3 Mechanics of wh-movement

We now turn to how wh-questions are derived syntactically. Under our approach to θ -role assignment, θ -roles are determined by the UTAH which essentially inspects the initial Merge of the DP argument with the verb (or other major category). Take our initial example again:

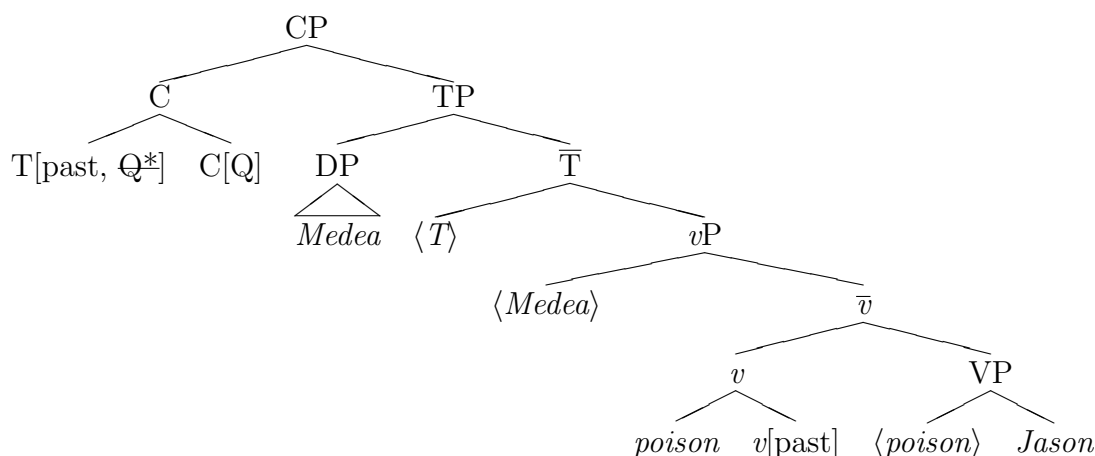
(49) Who did Medea poison?

Since in a wh-question like (49), it is the wh-expression that is interpreted as bearing the Theme θ -role, we assume, as discussed above, that this expression was originally Merged with V, and is then displaced to its subject position. The question then arises of what motivates this movement, and why it only occurs with wh-expressions.

9.3.1 Checking wh-features

Recall that, in our analysis of Subject Aux Inversion, we proposed that the feature [Q] was involved. We argued that English had a complementizer marked with [Q] which caused the sentence to be interpreted as a question. We then proposed that a strong version of Q appeared as the value of a [*u*clause-type] feature on T. When C[Q] values [*u*clause-type] on T as [Q*], T must move into a local configuration with C[Q]. Since T is a head, rather than a phrase, it raises and head-adjoins to C. This gives us the following structure:

(50)



Looking at wh-questions, we see that T to C movement takes place here too, and that the wh-expression moves into the specifier of T. We can implement this by assuming that C[Q] can optionally bear a strong [*uwh**] feature which checks with a wh-expression it c-commands. Since [*uwh**] on C is strong, the wh-expression must move into a local configuration with C. Since the wh-expression is a phrase, it moves into the specifier of CP. Roughly, we have:

(51) $[XP[wh] \ C[Q, \text{ } \cancel{uwh}^*] \ \dots \langle XP[wh] \rangle]$

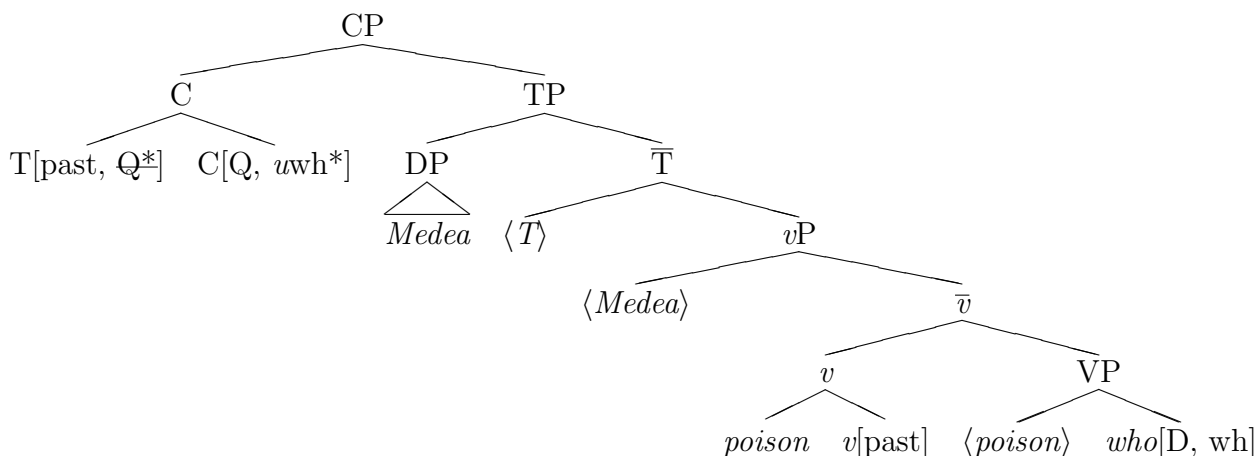
We treat this $[uwh^*]$ feature as being optional, since in a yes/no question, no *wh*-expression appears to the left of the inverted auxiliary:

(52) Did Medea poison Jason?

The existence of these sentences suggests that $[uwh^*]$ on $C[Q]$ is optional, since there is no *wh*-expression in this sentence. we will, however, return to hwather this is the right analysis below.

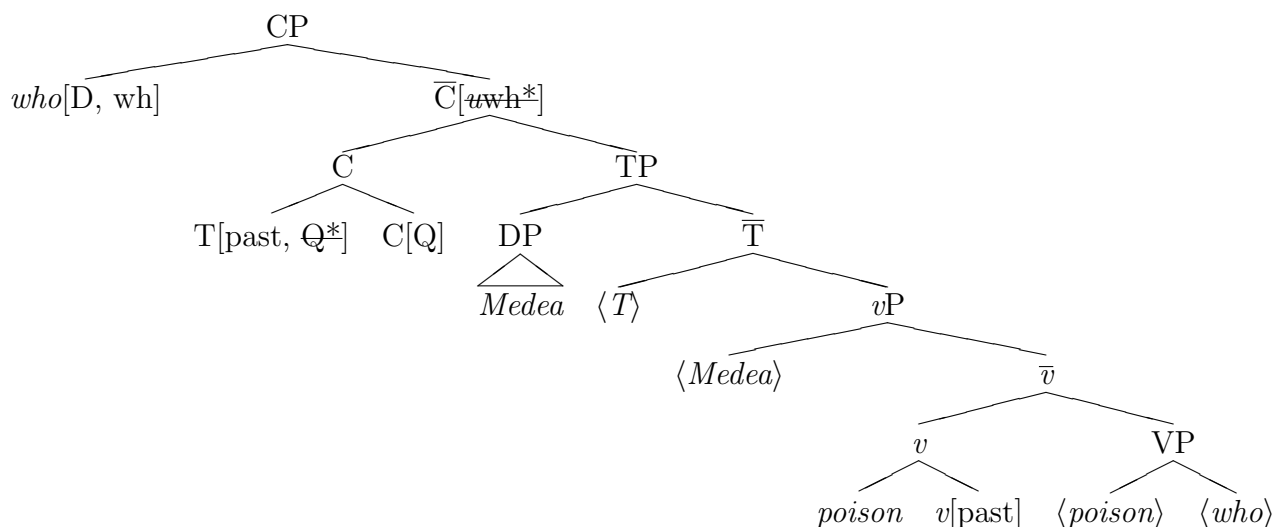
To see this proposal in action, consider a stage in a derivation just like that in (52) above, except that the object is a *wh*-expression, and C bears $[uwh^*]$:

(53)



The complementizer $C[Q, uwh^*]$ checks its $[wh]$ feature with *who* $[wh]$ under c-command (Agree), the strength of this feature forcing surface locality to obtain between C and *who*. Since *who* is phrasal, it moves into the specifier of C :

(54)



Technically, the locality enforced by the strength of $[uwh^*]$ is achieved by projecting this feature to the $\bar{C}$ level, just as the EPP feature is projected to $\bar{T}$, or the $[gen^*]$ feature to $\bar{D}$. When the *wh*-expression Merges with $\bar{C}$, it is in a sisterhood relation with the strong *wh*-feature.

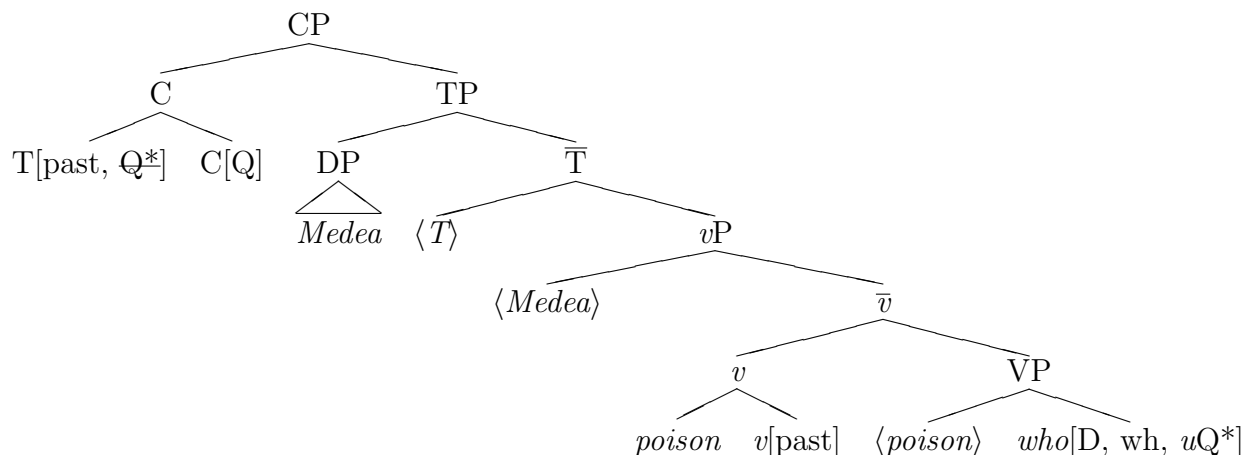
However, the optionality of the *wh*-feature on $C[Q]$ which we proposed to deal with the existence of simple yes/no questions actually leads to a serious problem. Let us assume that we decide to take the option not to specify $C[Q]$ with $[uwh^*]$. If we do this then, although we correctly predict that an auxiliary can move across a subject to C , we incorrectly predict that the following sentence should also be fine:

(55) *Did Medea poison who?

This is because *who*, so far, has no special requirement placed on it. It bears an interpretable *wh*-feature, which does not need to be checked. T to C movement has taken place to satisfy the strength of the strong $[Q^*]$ on T . The derivation should converge quite happily.

We could solve this problem by proposing that *wh*-expressions themselves have a strong feature (perhaps a Q feature) which forces them to move to the specifier of CP . This would give us a structure like (56):

(56)



Since, under this particular hypothesis, the $[uQ^*]$ feature on *who* is strong, then this wh-expression will have to move to ensure that it is in a local configuration with its checker. This approach would allow us to rule out (55), since in that example, the wh-expression *who* stays in its base position.

However, this doesn't seem to be the right way to go. Wh-expressions are perfectly well formed in sentences like the following, where they are apparently in their base position (such wh-expressions are commonly said to be **in situ**):

- (57) Medea saw who?
 (58) Jason gave the poisoned clothes to who?

These are called **echo-questions**, and they are well formed under a particular intonational contour, and under a particular interpretation where the paraphrase is not 'for which x ...' but rather 'repeat the word which *who* stands for in this sentence'. Echo-questions are usually used to express surprise or amazement, or to simply request that a part of a sentence should be repeated for clarity. They are not questions in the usual sense of the word, and don't seem to involve the kind of semantics we have been attributing the the feature $[Q]$ on C.

We conclude from this that if there is a C in an echo question, it does not bear $[Q]$. If we go down the route of attributing a $[uQ^*]$ feature to a wh-expression to make it move, then we would have to face the question of why wh-expressions do not seem to need to check this feature in echo-questions.

In addition, English, and many other languages, allow **multiple wh-questions**, where there is more than one wh-expression in the clause:

- (59) Who poisoned who?
 (60) Who showed what to who?

These can have the typical wh-quantifier interpretation, with rough paraphrases as follows:

- (61) For which pair(s) x and y, did x poison y?
 (62) For which triple(s) x, y and z, did x show y to z?

These multiple wh-questions clearly have at least one wh-expression *in situ*, so an approach which requires all wh-expressions to move to some position to check a strong feature is problematic.

This discussion leaves us with our initial problem: how do we rule out cases where we have an unmoved wh-expression but where T has moved to C:

(63) *Did Medea poison who?

Let us summarize where we have got to.

- We have motivated the idea of a wh-feature on the basis of morphological and semantic evidence.
- We can implement wh movement by assuming that C[Q] can optionally bear a strong [*uwh**] feature. The presence of this feature means that a phrase bearing an interpretable [wh] will have to move to the specifier of CP.

However,

- We found that our system as it stands has no way of ruling out examples where we have T to C movement but no wh-movement. This is because we have treated the [*uwh**] feature on C[Q] as optional.

9.3.2 Refining the system

In this section we will tackle the problem we outlined above. This problem clearly stems from the idea that the [*uwh**] feature on C[Q] is optional. What happens, then, if we make it obligatory?

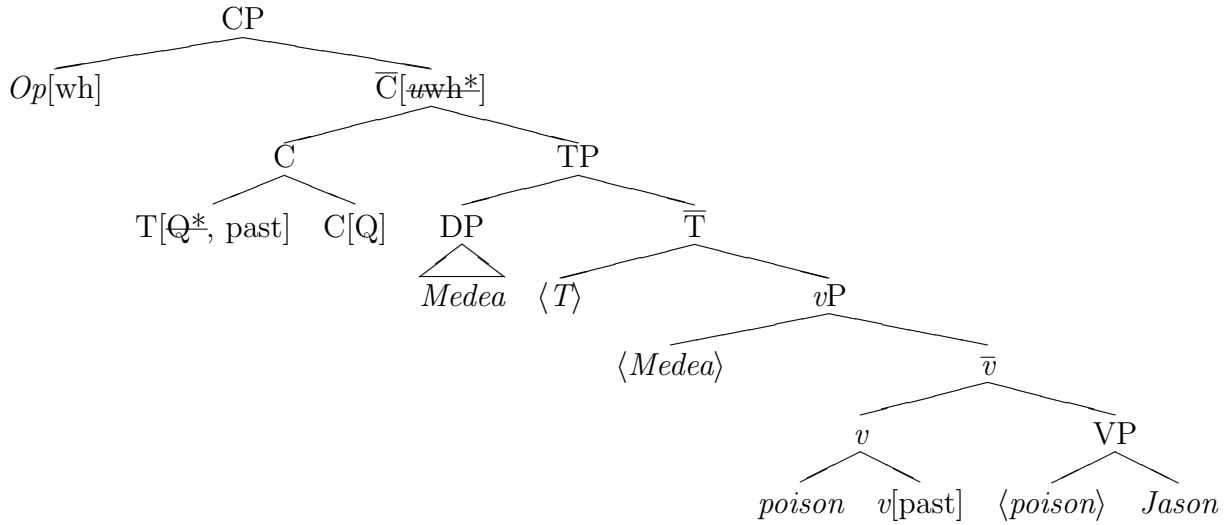
The obvious problem occurs with yes/no questions. These are questions, and so must have a C[Q], but there appears to be no wh-element, as we saw above:

(64) Did Medea poison Jason?

If C[Q] always bears [*uwh**], then how do we explain this?

We can solve this problem by assuming that the obligatory [*uwh**] feature on C[Q] is satisfied in a yes/no question by a null expression bearing a wh-feature which is Merged directly in the specifier of CP. Following tradition, we will call this expression a **null operator** and notate it as *Op*:

(65)

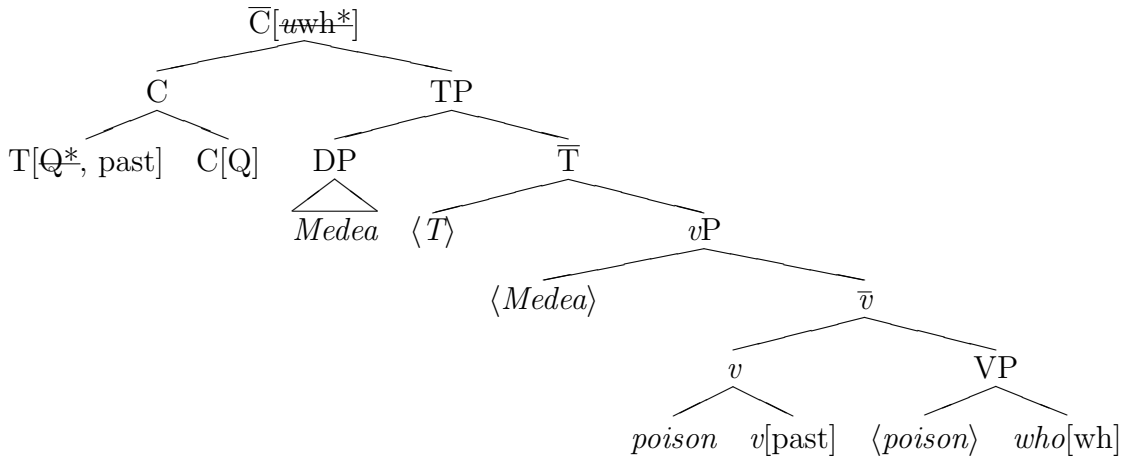


Once we have this idea in place, we can appeal to the preference for Merge over Move that we discussed in the last chapter to explain why (66) is ungrammatical:

(66) *Did Medea poison who?

Consider again a point in the derivation where T has been raised to $C[Q, uwh^*]$. We project the wh-feature to $\bar{C}$:

(67)



Now, assume that the numeration contains a null operator of the sort we just discussed (if it does not, then clearly the wh-expression *who* will have to raise to satisfy the strength of $[uwh^*]$). We now have a choice of operations which will satisfy the strength of the uninterpretable $[wh]$ feature on C: we can raise *who* or we can Merge the null operator. The preference for Merge over Move forces us to Merge the null operator.

However, the null operator does not have the semantics of an interrogative quantifier. Rather, it simply asks whether the proposition expressed by the sentence is true (in which case the answer is yes) or false (no). We can represent this semantics roughly as follows:

(68) Is it true or false that p?

Now, recall that for multiple wh-questions, we essentially constructed a complex interrogative quantifier, so that, for example, (69) has the reading in (70):

(69) Who kissed who?

(70) For which pair of entities *x* and *y* is it the case that *x* kissed *y*?

This semantics is triggered when there is more than one wh-expression in the sentence. Now, in the structure we have built up so far for our problematic example, we have two wh-expressions. One is the null operator, and the other is the expression *who*. Schematically, this looks as follows:

(71) *Op*[wh] ... *who*[wh]

If we try to apply the same semantic procedure, we find that we can't. Our procedure constructs a complex interrogative quantifier out of two simple ones. But in the case at hand, we don't even have two simple interrogative quantifiers: we have one (*who*) and an operator which gives rise to a yes/no question. Our semantic rule simply can't apply to the structure we have generated, and it is ruled out as perfectly well-formed syntax, but semantic gibberish. The final result is something which the semantic component just can't make any sense of.

The position we have taken, then, crucially appeals to the way that the semantic contribution of the syntactic structures is built up. A wh-expression may check the [*uwh**] feature in C syntactically. When the semantics comes to interpret the syntactic structure, it treats wh-expressions as wh-quantifiers. However, the null operator in yes/no questions requires a completely different kind of answer to other wh-quantifiers, which will lead to semantic incoherence if they are combined.

This is the position we will take: the [*uwh**] feature appears obligatorily on C[Q]. This feature is satisfied by movement of a wh-expression into the specifier of CP. In the absence of such a wh-expression, the derivation will have to contain a wh-operator which will give rise to a yes/no question semantics.

9.3.3 Other wh-movements

Before moving on to look at embedded questions and subject questions, we should note that wh-questions can also be formed from objects of prepositions and adjuncts:

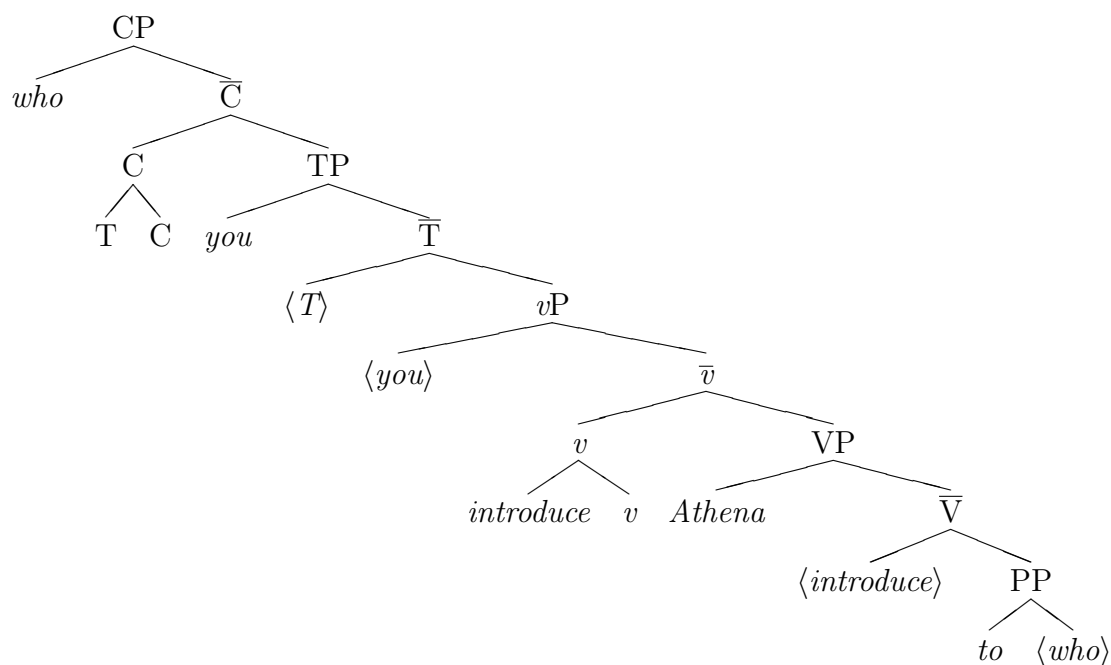
(72) Who did you introduce Athena to?

(73) Why did you kill Pegasus?

(74) Where did Perseus see the Gorgon?

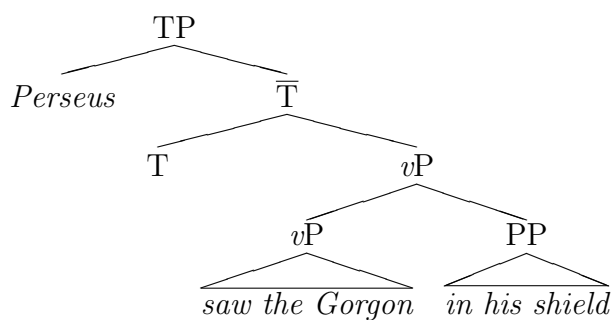
Analogously to our discussion of wh-objects, each of these wh-expressions is Merged in its base position and then moved. The object of the preposition is straightforward: it is initially Merged with the preposition, which it itself Merged with a projection of the verb. Later on in the derivation, the wh-expression is moved to the specifier of CP.

(75)



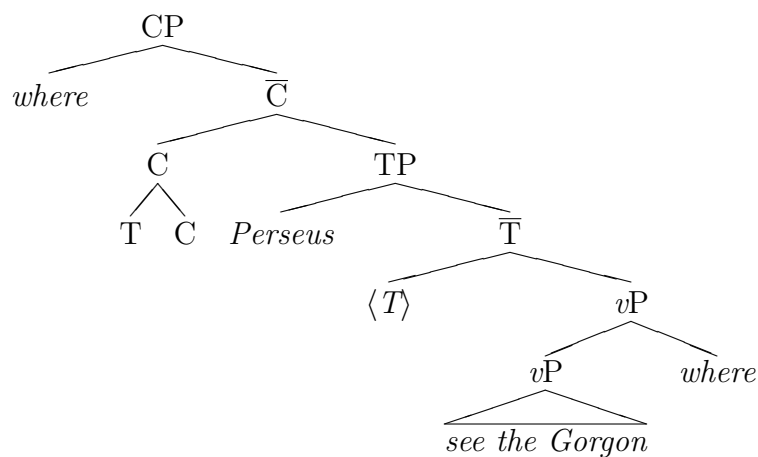
The adjunct cases work in just the same way. We have treated temporal, manner and causal adjuncts as adjoining to vP :

- (76) Perseus saw the Gorgon in his shield.



In this case, the PP may be a *wh*-expression *where* which moves to the specifier of CP:

- (77)



- (78) Where did Perseus see the Gorgon?

9.3.4 Embedded wh-questions

We now turn to wh-questions in embedded contexts. Recall that we saw yes/no questions could be embedded under certain predicates:

- (79) I wondered whether Medea had fled.
 (80) I inquired if we could leave early.

These same predicates allow wh-questions to be embedded under them as well:

- (81) I wondered who Medea had poisoned.
 (82) I inquired when we could leave.

As discussed in section 9.1, these wh-questions involve movement of the wh-expression to the specifier of the embedded CP. This means that the embedded C must be $[Q, uwh^*]$, given our discussion in the last section.

Notice, however, that in Standard English (and many dialects), no T to C movement takes place in embedded questions:

- (83) *I inquired could we leave early.

We can straightforwardly analyse this as arising from the lexical specification of the embedded C, which would have the property that it values the $[uclause\text{-}type]$ feature of T as just weak $[Q]$ rather than strong Q . Since our analysis of embedded questions and our analysis of wh-questions is syntactically parallel, we correctly predict that we do not find inversion in embedded wh-questions.

- (84) *I wondered who had Medea poisoned.
 (85) *I inquired when could we leave.

We will return to the possibility of having embedded inversion in the exercise section, when we look at some dialectal variation in embedded wh-question formation.

9.4 Subject wh-questions

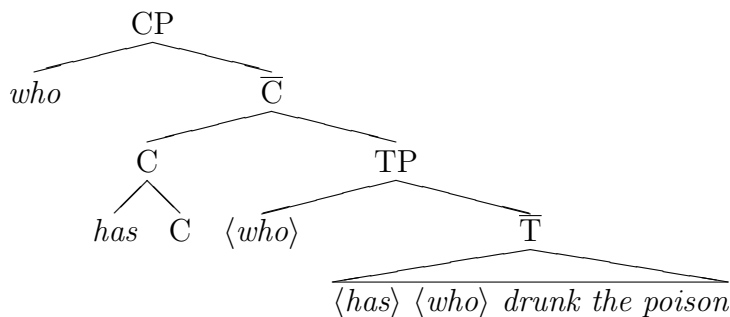
We now turn to subject wh-questions, which, in English, raise special problems. Take examples like the following:

- (86) Who has drunk the poison?
 (87) Which goddess might help us?
 (88) Who is sailing to Ithaca?

At first glance, it appears that we can treat these in a parallel fashion to the way that we dealt with object wh-questions. We derive a TP, as usual, with the wh-subject raising from

the specifier of vP (or from lower down in the case of unaccusatives) to the specifier of TP. We then Merge in $C[Q, uwh^*]$, which values $[uclause-type]$ on T as strong $[Q^*]$ causing T to raise to C, and then *who* to raise to the specifier of CP to check $[uwh^*]$, giving the following rough structure (I have suppressed irrelevant details in this case):

(89)



However, it turns out that this approach fails: we run into a problem with examples where there is no auxiliary:

(90) Who drank the poison?

(91) Which goddess helped us?

If the derivation we just went through is correct, then these sentences too should involve T to C movement. However, T to C movement where T contains just the T head with no attached auxiliary, usually triggers *do* support:

(92) Did you drink the poison?

(93) Did Athena help us?

This is because movement of T to C breaks the chain between T and v . Once the chain is broken, the tense features on T cannot be pronounced on v , and therefore *do*-support applies (see chapter 5, section 5.5).

Since the analysis we have built up in this chapter requires T to C movement in matrix wh-questions, we would predict that *do*-support should apply here too, giving the following examples:

(94) Who did drink the poison?

(95) Which goddess did help us?

These sentences are grammatical, but only on a special emphatic interpretation. What is worrying is that there appears to be no way to predict the grammaticality of the examples without *do*-support (examples (90) and (91)), or the ungrammaticality of (94) and (95) without a special interpretation.

The problem that we are faced with here arises because the complementizer which forces wh-movement in matrix clauses also forces T to C movement. The latter movement is forced because $C[Q]$ values the $[uclause-type]$ feature on T as strong Q, thereby triggering

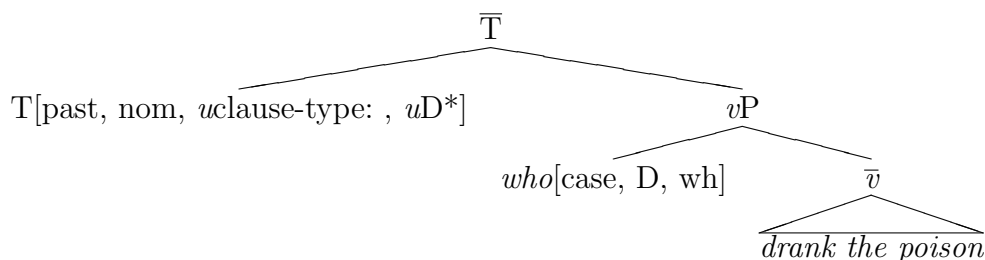
movement. The solution has to be something to do with the syntactic properties of *wh*-subjects in particular, since it is only here that the problem arises.

One crucial syntactic property of structural subjects which is not shared by objects and adjuncts is that the subject *c*-commands *T*. This opens up for us the possibility that it is actually the subject that checks the [*uclause-type*] feature on *T*, rather than *C*. Let us make this move, and see how successful it is:

(96) [*uclause-type*] on *T* may have [*wh*] as a value.

With this hypothesis in mind, now consider a point in a derivation where a *wh* subject is still *in situ* but *T* has been Merged:

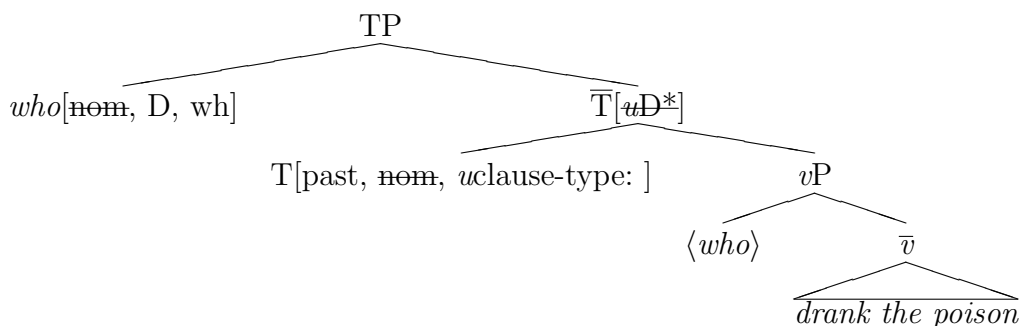
(97)



I have explicitly marked that *uclausetype* requires a value by placing a colon after it, rather than adopting our usual abbreviation. This is to make the details of the derivation as clear as possible.

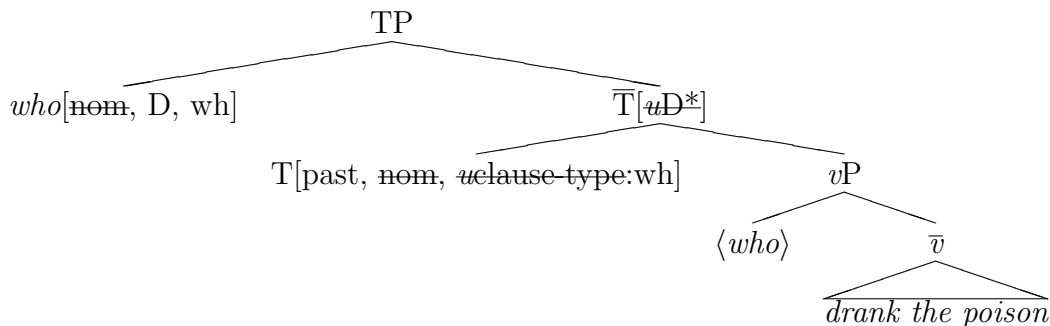
We now apply Agree between [*past*] on *T* and *uInfl* on *v*, valuing the latter so that the appropriate past tense form of *drink* eventually results. We also apply Agree between [*nom*] on *T* and [*case*] on *who*, valuing the latter, and checking both case features. We also apply Agree between EPP in *T* and the closest DP (which is *who*). To satisfy the strength property of EPP, we raise *who* to the specifier of *TP*:

(98)



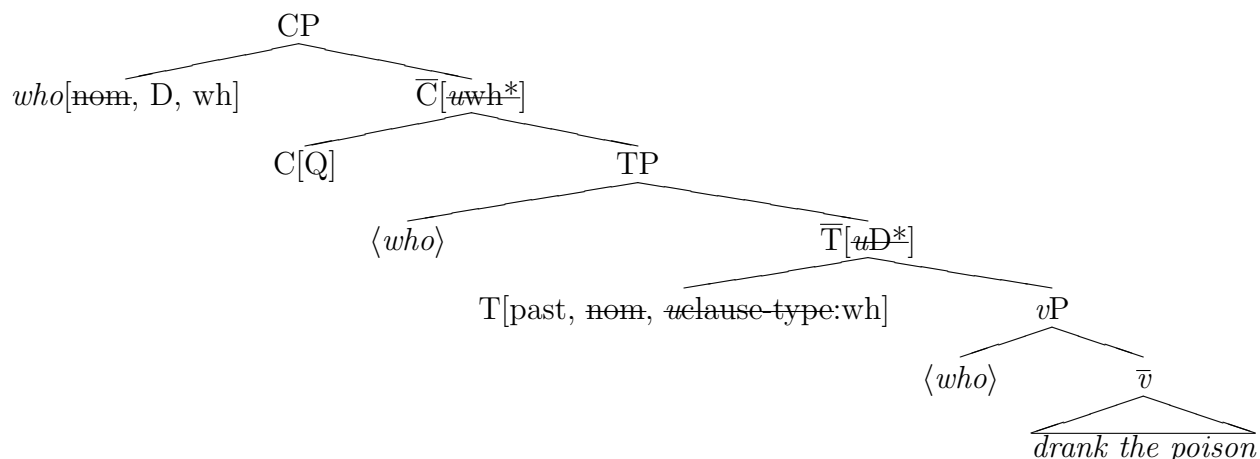
Now the subject DP *c*-commands *T*, and this DP bears an interpretable [*wh*] feature. Given our hypothesis about checking of *uclausetype*, we then apply Agree between *who* and *uclause-type* on *T*, valuing and checking this feature:

(99)



Finally, we Merge in $C[Q, uwh^*]$, in order to derive the fact that this is a question. The $[uclausetype]$ feature on T has been checked and valued, so Agree does not apply between C and T, and no movement of T to C takes place. The wh-subject, however, does raise to the specifier of CP, since Agree applies between it and $[uwh^*]$ on C:

(100)



If we look at this final tree, we see that all the relevant features have been checked, and that T has not raised to C as required.

9.5 Long distance wh-movement

Wh-questions have the interesting property that the wh-movement operation may apply across a clausal boundary:

(101) Who did Jason think (that) Medea had poisoned?

(102) What did you say (that) the poet had written?

In each of these examples, then wh element is assigned its θ -role by the verb in the embedded clause, but its surface position is the matrix specifier of CP.

There are two plausible hypotheses about how these examples are derived. The first idea is that the movement takes place in one step, so that the specifier of the embedded CP is never filled during the derivation. The other alternative is that movement is more

local, and that it takes place in two steps: firstly the wh-expression moves into the specifier of the embedded CP, and then it moves from this position to the specifier of the matrix CP. Schematically we have the two following derivations:

(103) XP[wh] ... think [_{CP} that ... ⟨XP[wh]⟩]

(104) XP[wh] ... think [_{CP} ⟨XP[wh]⟩ that ... ⟨XP[wh]⟩]

Simplicity might seem to favour the first derivation, in a number of ways. Firstly, it involves fewer operations, so it is more economical. Secondly, we know that verbs like *think* cannot take wh-question complements, and so it would seem sensible that they should never have such a complement during the course of a derivation. In other words, the existence of (105), would suggest that a configuration like (106) is illicit. Yet (106) is exactly the configuration which appears in a derivation like that sketched in (104):

(105) *Jason thinks who Medea had poisoned.

(106) ... think [[Wh-XP] ...

However, there is empirical evidence from various languages which suggests that the derivation involving movement to the intermediate specifier is the correct one. These are languages where these **long-distance questions** involve a special intermediate complementizer.

One good example of such a language is Scottish Gaelic. In this language the normal embedding complementizer is *gu*(*n*) (the appearance of the final *n* here is a purely morphophonological fact, and irrelevant to the syntax):

(107) Bha mi ag ràdh **gun** do bhuail i e
Was I asp saying that prt struck she him
'I was saying that she hit him.'

(108) Tha mi a' smaoin eachadh **gu** bheil Iain air a mhisg
Am I asp thinking that is Iain in his drink
'I think that Iain is drunk.'

When we form a wh-question on one of the arguments of the lower clause, we see that the intermediate complementizer changes its form from *gu* to *a* (with a concomittant change in the following verb form):

(109) Cò bha thu ag ràdh **a** bhuail i
Who were you asp saying that struck she
'Who were you saying that she hit.'

(110) Cò tha thu a' smaoin eachadh **a** tha air a mhisg
Who are you asp thinking that is on his drink
'Who do you think is drunk.'

On the hypothesis that movement takes place in one single swoop, from the θ -position to the matrix specifier of CP, there seems to be no reason for the change in complementizer. However, if the wh-expression moves into the specifier of the intermediate complementizer, then this gives us a plausible reason for the change. We can say that the intermediate com-

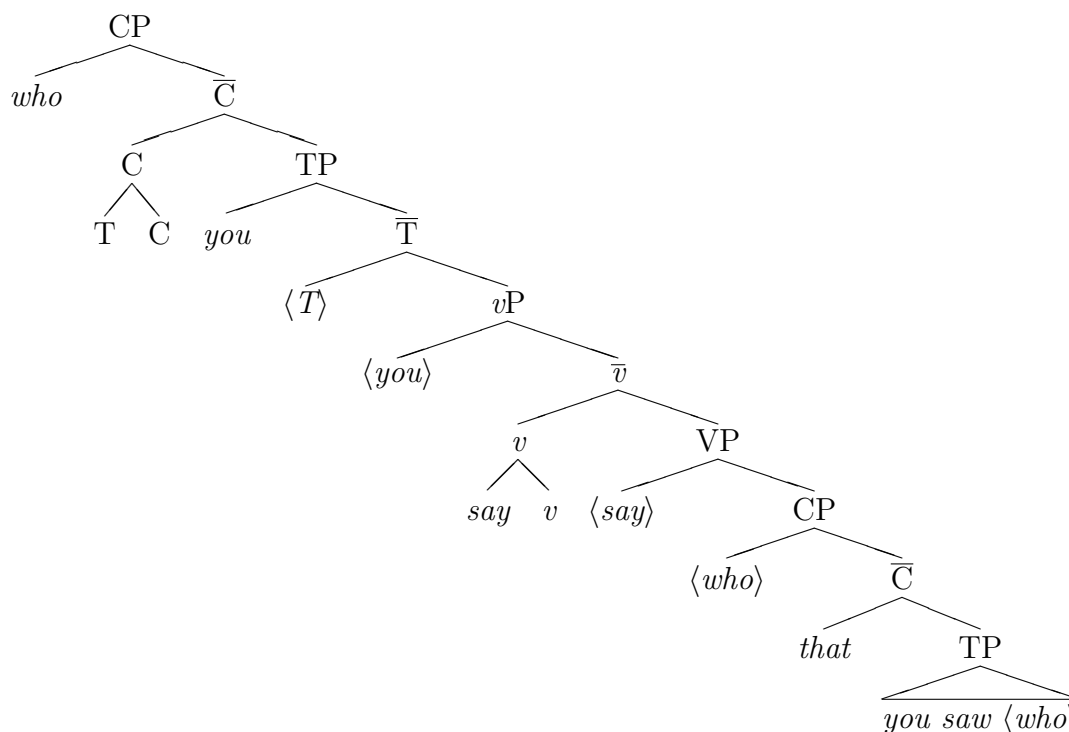
plementizer bears a strong $[uwh^*]$ feature, which is checked off by moving the wh-expression into its specifier. The complementizer $gu(n)$ does not bear this feature.

Note that this is motivation for the idea that wh-movement, after its initial launch from its Merged position, proceeds from specifier of CP to specifier of CP, and that it cannot skip a specifier of CP. It does not, on its own, derive this restriction. For the moment we will just accept this as a stipulation, which we will return to in the next chapter:

- (111) A wh-expression cannot skip a specifier of CP when it moves.

This idea forces us to adopt derivations like the following for long distance wh-questions:

- (112)



We will return in the next chapter to many more cases which argue for this kind of derivation.

9.6 Wh-in-situ

We discussed cases of **wh-in-situ** earlier on in this chapter. We saw two such cases: echo-questions and multiple wh-questions:

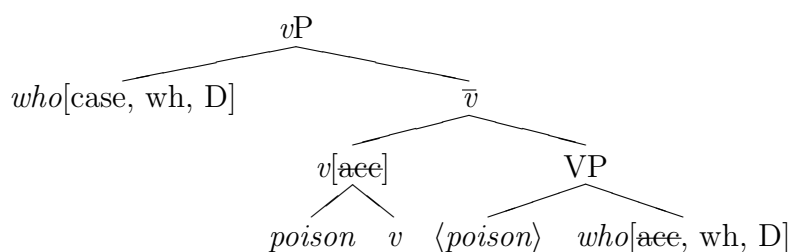
- (113) Medea poisoned who?

- (114) Who poisoned who?

Whereas (113) does not have the semantics of a true question, (114) clearly does. Let us see how it is derived.

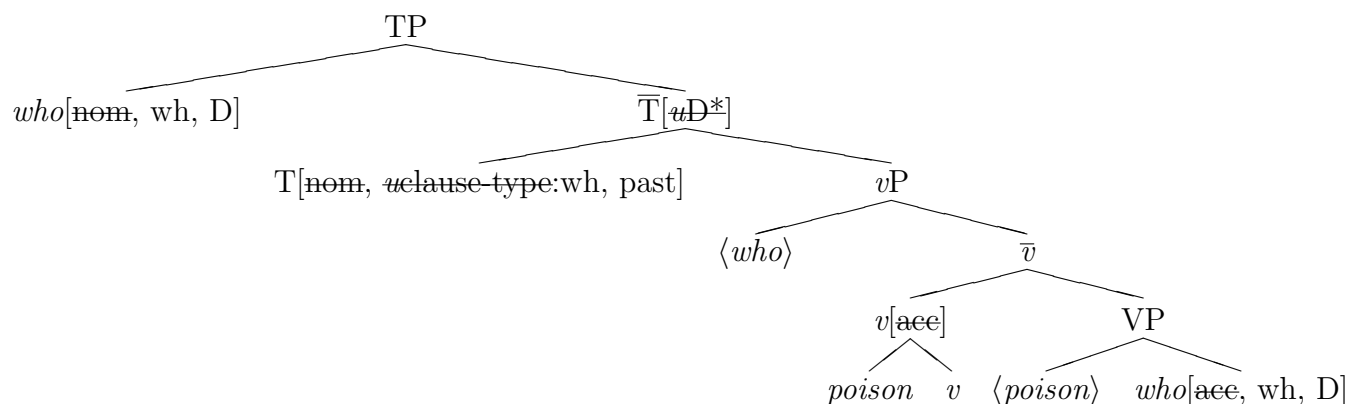
Recall that *wh* features on a *wh*-expression are interpretable. It is the strength of the uninterpretable *wh*-feature on *C* which causes *wh*-movement to take place. In an example like (114), we first Merge *who*[acc] with the verb, then having built up the *vP*, we have (115):

(115)



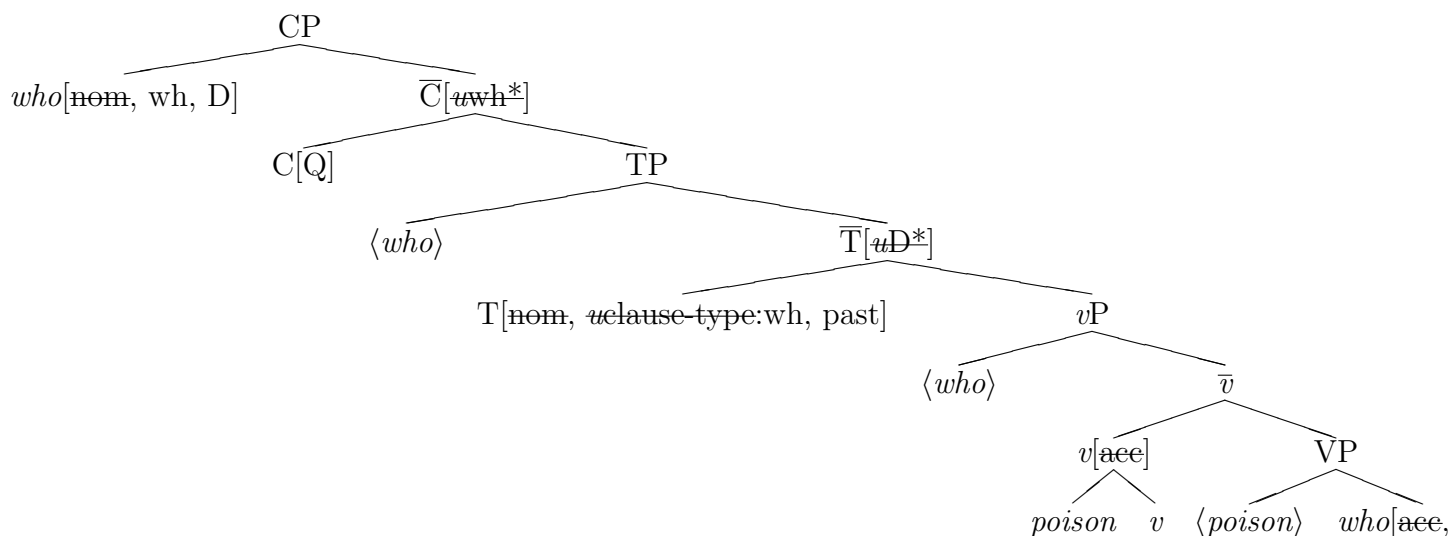
We then Merge *T*, checking case and attracting the subject which checks the EPP feature. Once the subject has raised, it Agrees with the [*u*clause-type] feature on *T*, valuing it:

(116)



Finally, we Merge in *C*[Q, *uwh**], which triggers movement of the *wh*-phrase in subject position to the specifier of *CP*, checking off the uninterpretable features on *C*:

(117)



Notice that all the uninterpretable features have been checked in this final syntactic structure to which the semantic rules can then apply. We therefore derive (118), with the rough interpretation in (119):

(118) who poisoned who?

(119) For which pair x and y is it the case that x poisoned y .

9.7 Superiority

Notice that we also immediately rule out a derivation where the *wh*-object Moves across the subject, deriving (120):

(120) * $[_{CP}$ Who $[_{TP}$ who poisoned]]

In the derivation for this sentence, we have raised the *wh*-subject to the specifier of TP, valuing $[uclause\text{-}type]$ on T, so no T to C movement takes place. We have then raised the object across the subject and Merged it in the specifier of CP, thereby checking the strong $[uwh^*]$ feature of C[Q].

In fact, our general approach to feature checking already rules (120) out. Remember that a head checks the closest matching feature that it c-commands. In the present case, when C is Merged with the TP, the closest *wh*-phrase is the one in subject position (this intervenes between C and the *wh*-object), so it is this one that raises into the specifier of CP. This effect is traditionally known as a **superiority effect**, since it is the superior (or highest) *wh*-phrase that moves.

Superiority also surfaces with three argument verbs. In an example like (121), the Theme c-commands the Goal, according to our analysis from Chapter 4:

(121) Athena introduced [Medea [to Jason]]

If we replace both the Theme and the Goal by wh-phrases, then we predict that it is the Theme, rather than the Goal, that undergoes wh-movement, since it is closest to the complementizer. This is the correct prediction:

(122) Who did Athena introduce to whom?

(123) *Who did Athena introduce who to?

We should note at this point, though, that superiority appears to be a weak phenomenon. If we replace simple wh-words with a more complex expression containing *which*, then both orders of wh-expression are well formed:

(124) Which poet wrote which ode?

(125) Which ode did which poet write?

This effect is known as the **D(iscourse)-linking effect**, since it occurs when the wh-determiner ranges over elements which are already established in the discourse. Notice that it is infelicitous to use a wh-phrase headed by *which* unless the discourse has already established the existence of the things that *which* is asking a question about:

(126) Look at all these books. Which book would you like?

Non-D-linked wh-words, like *who*, are less fussy about what has been mentioned in the previous discourse. It's perfectly fine to open a discourse with a wh-question containing one of these words. Imagine, for example, a context where you have entered a dark room and you hear a noise, it's fine to say (127), and distinctly odd to say (128):

(127) Who's there?

(128) # Which person is there?

We shall not attempt to provide an explanation for the D-linking effect in this book. See the reference section for further reading.

9.8 Cross-linguistic variation in wh-constructions

Our approach to wh-movement relies on the idea that it is motivated by a strong [*uwh**] feature on C. Given our previous approach to cross-linguistic variation, which ties most variability down to the strength properties of features of functional heads, we predict that there should be languages which have a weak [*uwh*] feature on C[Q]. It would follow that these languages should have wh-in-situ generally, even in simple wh-questions. There are, in fact, many languages with this property, among them Chinese and Japanese.

Compare the example below from English with its Japanese counterpart:

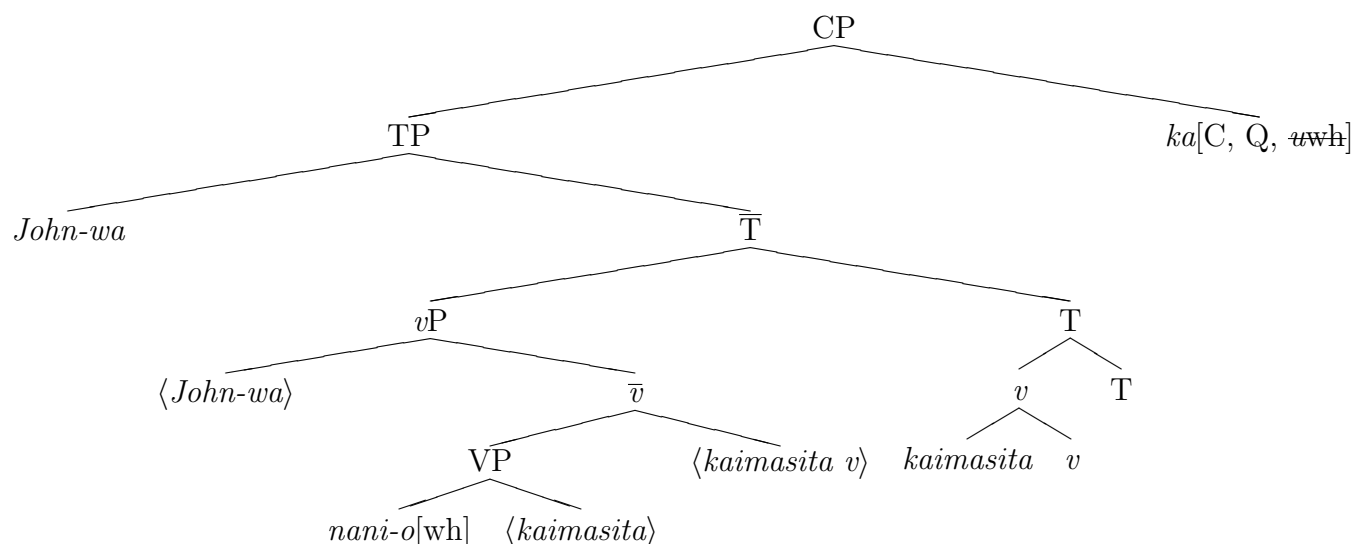
(129) What did John buy?

(130) John-wa nani-o kaimasita ka?
 John-TOP what-ACC bought Q
 'What did John buy?'

Japanese is a head final language, so the verb occurs to the right of its complement, and the complementizer (*ka*) appears to the right of the inflected verb. The *wh*-element is accusative, and appears in just the normal position for an accusative object. In other words, it does not undergo *wh*-movement.

Roughly, we have the following structure:

(131)



Here the $[uwh]$ feature on $C[Q]$ is weak, and so it is checked by being in an Agree relationship with the *wh*-object, but no movement takes place. Chinese works in just the same way.

There are also languages which move all their *wh*-phrases to some position which is peripheral in the clause. Here is an example from Bulgarian:

- (132) Kogo kakvo e pital Ivan
 Whom what Aux asked Ivan
 ‘Who did Ivan ask what?’

This is actually somewhat unexpected under our current assumptions, since once one *wh*-expression has moved to the specifier of CP, it should check a strong *wh*-feature there, and the derivation should converge, with all other *wh*-phrases in situ. Clearly something extra is needed to capture such languages. We will not delve into this problem here, but see the reference section for further reading.

We can now enter this parameter into our table, and we find the following set of properties:

(133)

	tense on Aux	tense on <i>v</i>	EPP on T	Decl on T	[top] on C	wh on C
English	strong	weak	strong	weak	optional	strong
French	strong	strong	strong	weak	?	optional
Swedish	weak	weak	strong	strong	strong	strong
Scottish Gaelic	strong	strong	weak	weak	weak	strong
German	strong	strong	strong	strong	strong	strong
Japanese	strong	strong	strong	weak	weak	weak

Japanese seems to have V to T movement, hence the [strong] specification for the Infl properties of Aux and *v*, although, as with many head final languages, the jury is somewhat out on this. There is also some evidence that there is a *v*P-external designated subject position in Japanese, hence the [strong] EPP feature. There is no evidence for V2, and so Japanese contrasts with German in specifications for Decl and [top], and since it is wh-in-situ, it has a weak value for [wh] on C. All of the other languages require the specifier of C in a wh question to be filled by a wh-phrase, hence their strong features for this parameter, except French, which appears to allow both possibilities.

Of course, as we expand this table, the particular specifications of parameters become more and more controversial, and few of these parameters have been settled decisively. However, the growing typology should give you an idea of how cross linguistic effects can be dealt with in the Principles and Parameters model we are developing here.

9.9 Summary

In this chapter we have looked at wh-movement, and argued that it was driven by strong wh-features which appear on C. We have argued that simple yes/no questions also involve a wh-expression which is Merged directly in the specifier of CP.

We have not introduced many new aspects of the theoretical system in this chapter. rather we have explored how the system which we set up to deal with the syntax of DP-movement can be extended to wh-movement. By and large, this extension has been successful, with only a single stipulation needed to deal with the lack of T to C movement in subject wh-questions.

Not only do the mechanics of the system work for simple wh-questions, they extend naturally to wh-in-situ phenomena, both in English and in other languages. One interesting prediction which is at least partially correct is the way that wh-movement works when there is more than one wh-expression in the sentence. we saw that (putting the D-linking phenomenon aside), the system we adopted for feature matching, which was independently motivated for dealing with DP-movement, also captured the superiority effect both in subject-object multiple wh-questions, and in Theme-Goal ones.

We have also introduced the idea that wh-movement, when it is cross clausal, always stops off in the intermediate specifier of CP, if there is one. We will explore the ramifications of this idea, and the motivations behind it, in much more detail in the next chapter.

Finally, we looked very briefly at how wh-constructions vary cross-linguistically, as usual concentrating on what predictions varying the strength of features makes. We saw that our system correctly predicted that some languages should have generalized wh-in-situ.

Phenomenon	Explanation
Wh-questions with one wh-word	Movement of wh-phrase to the specifier of CP triggered by [<i>uwh</i> ∗]
Yes/No Questions	Null <i>Op</i> [wh] Merged in Specifier of CP
Subject wh-questions	[<i>uclause-type</i>] feature on T can have [wh] as a value, so no T to C takes place
Complementizer Alternations in Gaelic	Movement takes place from specifier to specifier
Multiple wh-questions	Only one wh-phrase needs to move to specifier of CP
Superiority	Locality condition on Agree applies to wh-feature matching
Wh-in-situ languages	weak [<i>uwh</i>] feature in C

Exercises

Exercise 1 — Wh and Binding

Since reflexives can appear in DPs, and be bound, and since DPs can happily wh-move, we expect to find reflexives in moved wh-phrases. This is exactly what we do find:

- (1) Which book about himself did Anson write?

Part A

Explain why (1) above is problematic for the rule of reflexive binding developed in the previous chapter, on the assumption that this rule applies to the final syntactic representation.

Now assume that the rule applies instead during the derivation, just as Agree usually does. Go through a derivation for (1), and explain what the result of this change is.

Part B

Interestingly, binding can give us evidence for the intermediate sites of wh-movement. Consider an example like (2):

- (2) Which book about herself did Jenny say that Anson had written.

For most speakers, (2) is perfectly grammatical. Go through a derivation of (2), showing at which point the reflexive rule has to apply, adopting the analysis we gave in the text where the wh-phrase stops off in the intermediate specifier of CP. Once you have done this, show why a derivation which moved the wh-phrase in one sweep, from its θ -position to the matrix specifier of CP, would predict(2) to be ungrammatical, contrary to fact.

Exercise 2 — On categories that undergo wh-movement.

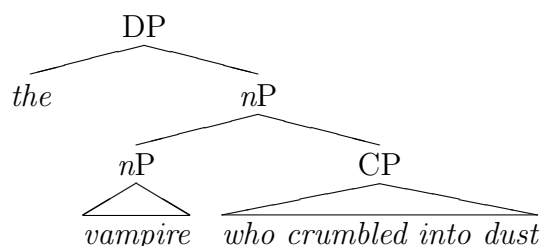
What do you think the category of the wh-phrase in (1) is, given the other examples? Justify your answer:

- (1) What did he reply?
- (2) He replied that he was happy.
- (3) *He replied his answer.

Exercise 3 — Relative Clauses

In the chapter we looked at wh-movement in questions. We also find wh-movement in Relative clauses. Relative clauses are clauses which can be used to modify nouns. We will assume that they are CPs attached at the *nP* level, just as other adjuncts are:

(1)



Here are some more examples.

- (2) The book [which Anson lent to Jenny]
- (3) The time [when you left me]
- (4) The place [where we lived for six years]
- (5) The vampire [who crumbled into dust]

Part A

Provide an argument, on the basis of the c-selectional requirements of elements in the sentences above (where possible), that relative clauses which have a wh-word in them involve movement. You will have to make up the relevant sentences.

Part B

We will assume that these relative clauses are just like wh-questions, except that their C which heads them bears an interpretable clause-type feature [Rel], rather than [Q]. This is why they are interpreted as relative modifiers rather than as questions. Given this assumption, draw trees for each of the DPs in the examples above, and specify whether [Rel] as a value of [*u*clause-type] on T is weak or strong.

Part C

This approach predicts that relative clauses should occur long distance. Is this true? Make up some relevant examples.

Part D

As well as relative clauses with movement of a wh-expression, we also find relative clauses which contain just the complementiser *that*:

- (6) The book [that Anson lent to Jenny]
- (7) The time [that you left me]
- (8) The place [that we lived in for six years]
- (9) The vampire [that crumbled into dust]

What problems arise in carrying over the analysis in part A to these examples? Can you think of any ways around these problems?

Exercise 4 — Negative movement

The following sentences involve movement of a negative phrase. Some of them are only well formed in a rather high register of English. Compare these examples with parallel wh-movement examples, and hypothesize a derivation for each example:

- (10) Never will I do syntax again.
- (11) Nothing like that would I ever eat again.
- (12) Nobody (*did) put up with her.

Once you have done this, look at the following example, and explain whether it is problematic for your approach, or whether it follows from it.

- (13) I swore that never again would I drink absinthe.

Exercise 5 — Trees

Provide tree structures for the following sentences, with a commentary on how the checking of features works.

- (1) Why did you say that you were leaving?
- (2) I asked who John would introduce to who.
- (3) Which book about Ulysses did you say that you would read?
- (4) Who did you say that John thought would leave early?
- (5) Who has drunk my whiskey?
- (6) Who seemed to have left first?

- (7) Whose poem about Achilles did Homer persuade Jason that he should read?
- (8) Who did Gilgamesh believe to have kissed Enkidu?

Exercise 6 — Binding Again

In previous exercises we have developed a number of generalisations about the binding of reflexives and pronouns. The Binding Generalisations we saw in Chapter 4 are repeated here:

- (9)
 - a. A reflexive must be bound.
 - b. A pronoun cannot be bound.
 - c. An R-expression cannot be bound.

We saw in Chapter 8, exercise 10, that at least the condition on reflexives had to be further restricted, to rule out examples like (10):

- (10) *Anson thought that himself was going to the club.

Let us hypothesize that the locality domain that is relevant to binding is the CP, so that the generalizations now read as follows:

- (11)
 - a. A reflexive must be bound to an antecedent in the same CP.
 - b. A pronoun cannot be bound to an antecedent in the same CP.
 - c. An R-expression cannot be bound.

Use these new generalisations to discuss the binding possibilities for the pronouns and reflexives in the following sentences.

- (12) *Dracula thought that himself was the Prince of Darkness.
- (13) Dracula thought that he was the Prince of Darkness.
- (14) Dracula thought himself to be the Prince of Darkness.
- (15) Dracula thought him to be the Prince of Darkness.
- (16) He thought that Dracula was the Prince of Darkness.

Further Reading

Section 9.1: The phenomenon of wh-movement received initial extended theoretical attention after the publication of Chomsky (1977), although there were important studies before this date as well (Baker 1970). Chomsky has repeatedly returned to this construction, and related ones (see Chomsky 1981, Chomsky 1982, Chomsky 1986a, Chomsky and Lasnik 1993, Chomsky 1995b). See Emonds (1976) for a broad ranging discussion of English wh-constructions.

Section 9.3: The approach we adopt here is influenced by Rizzi (1996), and especially by Chomsky (2000), although I have adapted ideas from these papers to fit in with the system we have been developing in the book.

Section 9.4: Subject questions in English have been a perpetual empirical thorn in the analytical side of wh-constructions. See Koopman (1983), Rizzi (1996) and Pesetsky and Torrego (2001) for different approaches.

Section 9.5: Chomsky argued for the idea that wh-movement took place in short steps as early as Chomsky (1973). This argument was made on the basis of unifying some of the locality effects on wh-movement we will talk about in the next chapter, and it wasn't until later that simple morpho-syntactic evidence of the sort discussed in this section (and in the next chapter) became available. See May (1979) and Freidin (1978) for some discussion, and, more recently McCloskey (forthcoming) and Adger and Ramchand (2002).

Section 9.6: **Section 9.7:** The idea of superiority was first discussed by Chomsky (1973). See Pesetsky (1987) of the notion of D-linking.

Section 9.8: See Huang (1982) for an extended discussion of the syntax of wh-in-situ languages, and Cheng (1991) and Richards (2001) for more recent theoretical discussion of parametric variation in wh-constructions, and see Rudin (1988) for an in depth discussion of multiple wh-fronting languages.